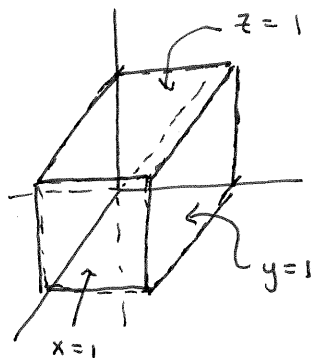


HOMEWORK 10 : Divergence Thm

Prob 1 :

(a) $\text{div } \vec{F} = 2x + 2y + 2z$

$$\begin{aligned}\iint_S \vec{F} \cdot d\vec{A} &= \iiint_T \text{div } \vec{F} \, dV \\&= \int_0^1 \int_0^1 \int_0^1 (2x + 2y + 2z) \, dV \\&= \int_0^1 \int_0^1 \left[x^2 + (2y + 2z)x \right]_0^1 dy \, dz \\&= \int_0^1 \int_0^1 (1 + 2y + 2z) \, dy \, dz \\&= \int_0^1 \left[y^2 + (1 + 2z)y \right]_0^1 dz \\&= \int_0^1 (1 + 1 + 2z) \, dz = \left[2z + z^2 \right]_0^1 = 3\end{aligned}$$



(b) $\text{div } \vec{F} = 3x^2 + 3y^2 + 3z^2$

Spherical Coordinates :

$$\begin{aligned}x &= \rho \sin \phi \cos \theta \\y &= \rho \sin \phi \sin \theta \\z &= \rho \cos \phi\end{aligned}$$

$$0 \leq \rho \leq 3$$

$$0 \leq \phi \leq \pi$$

$$0 \leq \theta \leq 2\pi$$

$$\begin{aligned}\iint_S \vec{F} \cdot d\vec{A} &= \int_0^{2\pi} \int_0^\pi \int_0^3 3\rho^2 \cdot \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \\&= \int_0^{2\pi} \int_0^\pi \left[\frac{3}{5} \rho^5 \sin \phi \right]_{\rho=0}^{\rho=3} d\phi \, d\theta \\&= 2\pi \cdot \frac{3^6}{5} \int_0^\pi \sin \phi \, d\phi = -2\pi \frac{3^6}{5} \cos \phi \Big|_0^\pi = 4\pi \cdot \frac{3^6}{5} = \frac{2916}{5} \pi \approx 1832.2\end{aligned}$$

(c) $\text{div } \vec{F} = 12x + 2y + 2$

Cylindrical Coordinates:

$$x = r \cos \theta$$

$$0 \leq r \leq 2$$

$$y = r \sin \theta$$

$$0 \leq \theta \leq 2\pi$$

$$z = z$$

$$0 \leq z \leq 3$$

$$\begin{aligned}\iint_S \vec{F} \cdot d\vec{A} &= \int_0^3 \int_0^{2\pi} \int_0^2 (12r \cos \theta + 2r \sin \theta + 2) r \, dr \, d\theta \, dz \\&= 3 \int_0^{2\pi} \left[4r^3 \cos \theta + \frac{2}{3} r^3 \sin \theta + 2 \frac{r^2}{2} \right]_0^2 d\theta \\&= 3 \int_0^{2\pi} \left(2^5 \cos \theta + \frac{2^4}{3} \sin \theta + 2^2 \right) d\theta \\&= 3 \left(2^5 \sin \theta - \frac{2^4}{3} \cos \theta + 2^2 \theta \right) \Big|_0^{2\pi} \\&= 3 \left(-\frac{2^4}{3} + 2^3 \pi \right) - 3 \left(-\frac{2^4}{3} \right) = 3 \cdot 2^3 \pi = 24\pi\end{aligned}$$

Problem 2: $\text{div } \vec{F} = 3$

$$\begin{aligned} \frac{1}{3} \iint_S \vec{F} \cdot \vec{n} dA &= \frac{1}{3} \iiint_D \text{div } \vec{F} dV = \frac{1}{3} \iiint_D 3 dV \\ &= \text{Volume}(D) \quad \checkmark \end{aligned}$$

Problem 3: $\text{div } \vec{F} = -2 - 8 + 10 = 0$

Let Φ_S denote the flux through surface S , then

$$\Phi_S = \Phi_A + \Phi_B + \Phi_C + \Phi_D + \Phi_{\text{Top}} + \Phi_{\text{Base}} \quad \text{WANT: } \Phi_{\text{Top}} = ?$$

$$\text{By the divergence thm: } \Phi_S = \iiint_T \text{div } \vec{F} dV = 0$$

For Φ_C : $x=0$ and $\vec{n} = [1, 0, 0]$

$$\begin{aligned} \Rightarrow \Phi_C &= \iint_C [-2x, -8y, 10z] \cdot [1, 0, 0] dA \\ &= \iint_C 2x dA = 0 \quad \text{since } x=0 \text{ on } C \end{aligned}$$

For Φ_D : $y=0$ and $\vec{n} = [0, -1, 0]$

$$\begin{aligned} \Rightarrow \Phi_D &= \iint_D [-2x, -8y, 10z] \cdot [0, -1, 0] dA \\ &= \iint_D 8y dA = 0 \quad \text{since } y=0 \text{ on } D \end{aligned}$$

For Φ_{Base} : $z=0$ and $\vec{n} = [0, 0, -1]$

$$\begin{aligned} \Rightarrow \Phi_{\text{Base}} &= \iint_{\text{Base}} [-2x, -8y, 10z] \cdot [0, 0, -1] dA \\ &= \iint_{\text{Base}} -10z dA = 0 \quad \text{since } z=0 \text{ on Base} \end{aligned}$$

We know: $\Phi_A = -4$ and $\Phi_B = -4$

Therefore,

$$\Phi_{\text{Top}} = \Phi_S - (\Phi_A + \Phi_B + \Phi_C + \Phi_D + \Phi_{\text{Base}}) = 0 - (-4 - 4 + 0 + 0 + 0) = 8$$

Problem 4: $\vec{F} = [C, D, E]$ where C, D, E are constant scalars.

By the divergence thm:

$$\iint_S \vec{F} \cdot d\vec{A} = \iiint_T \text{div } \vec{F} dV = 0 \quad \text{Since } \text{div } \vec{F} = 0$$